[536709] Stochastic Processes

Yuki Chino

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1 Preparation

1.1 Probability Space

We recall that probability is what we estimate the occurrence of some event. In words from the set theory, probability is

a mapping from the space of the events to an interval $[0, 1]$.

We call this mapping *probability measure* or *probability distribution*.

1.1.1 Measurable Space

Consider a coin toss. We have some primitive events 'Coin shows head' (we abbreviate by H), 'Coin shows tail ' (T)[1)]. We call them **sample** and the set of all samples **sample space**. In general we denote sample space by Ω.

We can define "event" as a set of samples, i.e. "event" is a subset of sample space Ω. When we estimate the occurrence of the event we interested in, it is important how we determine events we need to quantify their occurrence a priori.

[1)] Sometimes we have the situation 'coin stands', but we ignore it in general.

Now we have a question:

What kind of events should be sufficient to describe all events whose probabilities we want to assign?

Intuitively, σ-algebras specify which events are observable. The modern probability theory solved the question above to consider the following properties for "the space of events".

Definition 1.1. The space of events $\mathcal{F}$ satisfies the followings:

- $\Omega \in \mathcal{F}$
- $A \in \mathcal{F} \Longrightarrow A^c \in \mathcal{F}$
- $A_n \in \mathcal{F}\ (n \in \mathbb{N}) \Longrightarrow \bigcup_{n\in\mathbb{N}} A_n \in \mathcal{F}$

We call the space with the algebraic structure above **σ-algebra**. The σ-algebra defines the event mathematically. We call a pair $(\Omega, \mathcal{F})$ **measurable space**. The point is that three properties in σ-algebra guarantee measurability. In words, we can consider a measure on σ-algebra and measure each element, i.e. we can estimate the occurrence of each event with some measure.

1.1.2 Probability Measure

We call what we quantify the occurrence of each element in the space of events between $[0, 1]$ **probability**. We denote the mapping by $\mathbb{P} : \mathcal{F} \to [0, 1]$ and call it **probability measure** or **probability distribution**. The probability measure must be compatible with the σ-algebra structure.

Definition 1.2. The probability measure $\mathbb{P}$ satisfies the followings:

- For any $A \in \mathcal{F}$, $\mathbb{P}(A) \in [0, 1]$
- $\mathbb{P}(\Omega) = 1$
- $A_n \in \mathcal{F}\ (n \in \mathbb{N}), \quad A_i \cap A_j = \varnothing\ (i \neq j) \Rightarrow \mathbb{P}\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{n\in\mathbb{N}} \mathbb{P}(A_n)$

We sometimes call the second condition *σ-additivity* for measures. We define the **probability space** combining $\mathbb{P}$ with the measurable space $(\Omega, \mathcal{F})$. We denote the probability space by the triplet $(\Omega, \mathcal{F}, \mathcal{P})$. As a simple example we consider tossing a dice. If we consider $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $\mathcal{F} = \mathcal{P}(\Omega)$ and $\mathbb{P}(\{\text{The dice shows } i\}) = \frac{1}{6}$, then $(\Omega, \mathcal{F}, \mathbb{P})$is also a probability space. We call this distribution **uniform distribution**. The classical theory is mostly based on the uniform distribution.

When the sample space Ω is a topological space, the σ-algebra generated by family of all open sets is called **Borel set** of Ω and we denote it by $\mathcal{B}(\Omega)$. Especially, in the case $\Omega = \mathbb{R}^n$, we write $\mathcal{B}^n := \mathcal{B}(\mathbb{R}^n)$. The concept of Borel set is important to consider random variables defined on $\mathbb{R}^n$.

1.1.3 Distribution and Density

The probability theory is based on Measure Theory and Theory of Lebesgue Integration. We consider the relation between *cumulative distribution function* and (probability) *density function*. We first recall the definitions:

Definition 1.3. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and X be a real-valued random variable. We set

$$F_X(x) := \mathbb{P}(X \leq x). \tag{1.1}$$

If the distribution of X is absolutely continuous with respect to Lebesgue measure, then there exists a (probability) **density function** f_X such that

$$F_X(x) = \int_{-\infty}^{x} f_X(y)\, dy \qquad \text{for any } x \in \mathbb{R}. \tag{1.2}$$

F_X is called **cumulative distribution function** (c.d.f) of X and

From this definition, we can see the relation

$$f_X(x) = \frac{d}{dx} F_X(x), \tag{1.3}$$

where dx is the Lebesgue measure. More precisely, let μ be a reference measure, then the density f is defined as the Radon-Nikodym derivative:

$$f = \frac{d\mathbb{P}}{d\mu}, \tag{1.4}$$

which implies that for any measurable set $A \in \mathcal{F}$,

$$\mathbb{P}(X \in A) = \int_{X^{-1}(A)} d\mathbb{P} = \int_A f\, d\mu. \tag{1.5}$$

For the discrete case, we take the reference measure to be the counting measure.

Remark 1.4. In general we can not define the density for arbitrary measure μ. If we can define the density as in (1.4), then it is almost everywhere uniquely defined. □

1.2 Random Variable

For the first step of the modern probability theory we studied the probability space which corresponds to measure space in **Measure Theory** and **Theory of Lebesgue Integration**. In this section and the next section, we will review *random variable* and *expectation* corresponding to measurable function and Lebesgue integral, respectively.

1.2.1 Random Variable and Distribution Function

Roughly saying, the real-valued function X defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called *random variable* when it is $\mathcal{F}$-measurable.

Definition 1.5. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. We say a mapping $X \colon \Omega \to \mathbb{R}$ is **$\mathcal{F}$-measurable** if we have

$$\{X \leq x\} := \{\omega \in \Omega \colon X(\omega) \leq x\} \in \mathcal{F} \qquad \text{for any } x \in \mathbb{R}. \tag{1.6}$$

We call $\mathcal{F}$-measurable mapping $X \colon \Omega \to \mathbb{R}$ (real-valued) **random variable**. We also call

$$F_X(x) = \mathbb{P}(X \leq x) \tag{1.7}$$

cumulative distribution function of X.

A bit more general definition of random variable: for any $A \in \mathcal{B}$, X is called random variable if

$$X^{-1}(A) := \{\omega \in \Omega \colon X(\omega) \in A\} \in \mathcal{F} \tag{1.8}$$

We note here that **$\mathcal{F}$-measurability** guarantees a mapping (function) X on Ω can be estimated by a probability measure $\mathbb{P}$ on $\mathcal{F}$. When $\mathcal{F}$ is a Borel set, we call $\mathcal{F}$-measurable functions **Borel function**.

Definition 1.6. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. If a random variable X takes at most countable values, then we call it **discrete random variable**. In particular, in the case that the range of X is $\{x_n\}_{n\in\mathbb{N}}$, we call

$$p_X(x_n) = \mathbb{P}(X = x_n) \tag{1.9}$$

probability mass function of X.

Definition 1.7. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. If a random variable X is not discrete and we have the following expression with some function f_X

$$F_X(x) = \int_{-\infty}^{x} f_X(y)\, dy \qquad \text{for any } x \in \mathbb{R}, \tag{1.10}$$

then we call X **continuous random variable** and f_X is called (probability) **density function** of X.

1.3 Independence

When we have two more events or consider two or more random variables, we need to pay attention to the relation in between them.

Definition 1.8. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

(1) Let $A_1, A_2, \cdots \in \mathcal{F}$ be a sequence of events. We call the events $A_1, A_2, \cdots$ are **independent** each other when for arbitrary finite subsequence $A_{n_1}, A_{n_2}, \cdots, A_{n_k}$ $(k \geq 2, 1 \leq n_1 < \cdots < n_k)$, we have

$$\mathbb{P}\left(\bigcap_{i=1}^{k} A_{n_i}\right) = \prod_{i=1}^{k} \mathbb{P}(A_{n_i}). \tag{1.11}$$

If they are not independent, then we call they are **dependent**.

(2) Let $X_1, X_2, \cdots$ be a sequence of real-valued random variables on $\Omega, \mathcal{F}, \mathbb{P}$. We call the random variables $X_1, X_2, \cdots$ are **independent** each other when for arbitrary finite subsequence $X_{n_1}, X_{n_2}, \cdots, A_{n_k}$ $(k \geq 2, 1 \leq n_1 < \cdots < n_k)$, we have

$$\mathbb{P}\left(\bigcap_{i=1}^{k} \{X_{n_i} \leq x_{n_i}\}\right) = \prod_{i=1}^{k} F_{X_{n_j}}(x_{n_i}) \qquad \forall x_{n_1}, \cdots, x_{n_k} \in \mathbb{R}. \tag{1.12}$$

This definition is equivalent to independence of the σ-algebras generated by X_i. We recall here the definition of **disjoint**. Let A, B be events. We call events A and B are disjoint if we have

$$\mathbb{P}(A \cap B) = 0. \tag{1.13}$$

Example 1.9. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. We now consider a sequence of real-valued **independent identically distributed** (we normally write **i.i.d.**) Bernoulli random variables $X_1, X_2, \cdots, X_n$, i.e.

$$\mathbb{P}(X_1 = 1) = 1 - \mathbb{P}(X_1 = 0) = p \qquad (p \in (0, 1)). \tag{1.14}$$

If we define $Y = \sum_{k=1}^{n} X_k$, then Y is a binomial random variable with parameter (n, p). We consider the characteristic function of Y

$$\varphi_Y(\theta) = \mathbb{E}[e^{i\theta Y}] = \mathbb{E}\left[e^{i\theta \sum_{k=1}^{n} X_k}\right] = \prod_{k=1}^{n} \mathbb{E}[e^{i\theta X_k}] = \prod_{k=1}^{n} \mathbb{E}[e^{i\theta X_1}] = \Big(\mathbb{E}[e^{i\theta X_1}]\Big)^n. \tag{1.15}$$

Now we have $\varphi_{X_1}(\theta) = pe^{i\theta} + (1-p)$. Therefore,

$$\varphi_Y(\theta) = \Big(pe^{i\theta} + 1 - p\Big)^n, \tag{1.16}$$

which is the characteristic function of the binomial distribution with parameter (n, p). As a result by Glivenko's theorem (see Theorem 1.24), Y is a binomial random variable with parameter (n, p). ■

For $a, z \in \mathbb{R}^d$, we denote by $\langle a, z\rangle$ the scalar product of a and z, i.e.

$$\langle a, z\rangle = \sum_{i=1}^{d} z_i\, a_i \tag{1.17}$$

Definition 1.10 (Gaussian random variable)**.** If the characteristic function $\varphi_X(z)$ of $X = (X_1, X_2, \cdots, X_d)$ can be expressed by

$$\varphi_X(z) = \exp\Big(i\langle z, a\rangle - \frac{1}{2} z\, V\, z^{\mathsf{T}}\Big) \qquad z = (z_1, z_2, \cdots, z_d) \in \mathbb{R}^d \tag{1.18}$$

where $a \in \mathbb{R}^d$ and $V = (v_{ij})_{i,j=1,2,\cdots,d}$, then X is called d-dimensional **Gaussian random variable**. When $d = 1$, it is simply called **Gaussian**.

If V is positive definite, the distribution can be written by density function g:

$$g(x) = \big((2\pi)^d \det V\big)^{-1/2} \exp\Big(-\frac{1}{2}\langle (x-a)\, V^{-1}, \tag{1.19}$$

By differentiating the characteristic function $\varphi_X(z)$,

$$a_k = \mathbb{E}[X_k], \qquad v_{ij} = \mathrm{Cov}(X_i, X_j) \tag{1.20}$$

a is called **mean vector**, V is a **covariance matrix** and we denote by $N(a, V)$ the Gaussian.

Proposition 1.11 (Gaussian case)**.** *For a Gaussian system $X = (X_1, X_2, \cdots, X_d)$, the necessary and sufficient condition for the independence of $X_1, X_2, \cdots, X_d$ is*

$$\mathrm{Cov}(X_i, X_j) = 0 \qquad i \neq j, \tag{1.21}$$

that is, V is a diagonal matrix.

Proof. (Necessity) Since X_i and X_j are independent, we have

$$\mathrm{Cov}(X_i, X_j) = \mathbb{E}[(X_i - a_i)(X_j - a_j)] = \mathbb{E}[X_i - a_i]\,\mathbb{E}[X_j - a_j] = 0 \tag{1.22}$$

Therefore, V is a diagonal matrix.

(Sufficiency) When V is diagonal, we have

$$\varphi_X(z_1, z_2, \cdots, z_d) = \exp\left(\sum_{k=1}^{d} i a_k z_k - \frac{v_k}{2} z_k^2\right) = \prod_{k=1}^{d} \exp\left(i a_k z_k - \frac{v_k}{2} z_k^2\right) = \prod_{k=1}^{d} \varphi_X(z_k), \tag{1.23}$$

which shows the independence of $X_1, X_2, \cdots, X_d$. ■

When the covariance of X and Y is zero, we say that X and Y have **no correlation**. If X and Y are independent, then they have no correlation.

Example 1.12. Let X be a Gaussian random variable and set

$$Y = X^2 - 1 \tag{1.24}$$

Then since $\mathbb{E}[X^3] = \mathbb{E}[X] = 0$,

$$\mathrm{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X] = 0 \tag{1.25}$$

On the other hand, since $\mathbb{E}[X^4] = 3$ and $\mathbb{E}[X^2] = 1$,

$$\mathbb{E}[X^2 Y] = \mathbb{E}[X^4] - \mathbb{E}[X^2] = 2, \qquad \mathbb{E}[X^2]\,\mathbb{E}[Y] = 0. \tag{1.26}$$

Therefore, since $\mathbb{E}[X^2 Y] \neq \mathbb{E}[X^2]\,\mathbb{E}[Y]$, X and Y are not independent. ■

1.3.1 Conditional Probability

Definition 1.13. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $A, B \in \mathcal{F}$ be events satisfying $\mathbb{P}(A) > 0$. We define **conditional probability** under the event A as

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)}. \tag{1.27}$$

We note here that if A and B are independent, then since $\mathbb{P}(B \cap A) = \mathbb{P}(B)\mathbb{P}(A)$, we have $\mathbb{P}(B \mid A) = \mathbb{P}(B)$. This means that the event A never affects the occurrence of the event B.

Theorem 1.14. (Law of multiplication) *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For the sequences of events $A_1, A_2, \cdots, A_n \in \mathcal{F}$, if*

$$\mathbb{P}\left(\bigcap_{j=1}^{n-1} A_j\right) > 0, \tag{1.28}$$

then we have

$$\mathbb{P}\left(\bigcap_{j=1}^{n} A_j\right) = \mathbb{P}(A_1)\mathbb{P}(A_2|A_1)\mathbb{P}(A_3|A_1 \cap A_2)\cdots\mathbb{P}\left(A_n \Big| \bigcap_{j=1}^{n-1} A_j\right). \tag{1.29}$$

Theorem 1.15. (Total probability formula) *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For a disjoint sequence of events $A_1, A_2, \cdots, A_n \in \mathcal{F}$ ($A_i \cap A_j = \varnothing$, $i \neq j$), if*

$$\mathbb{P}(A_i) > 0 \quad (i = 1, 2, \cdots, n), \qquad \bigcup_{j=1}^{n} A_j = \Omega, \tag{1.30}$$

then we have

$$\mathbb{P}(B) = \sum_{i=1}^{n} \mathbb{P}(A_i)\,\mathbb{P}(B|A_i). \tag{1.31}$$

1.3.2 Conditional Expectation

Definition 1.16. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $A \in \mathcal{F}$ be an event satisfying $\mathbb{P}(A) > 0$ and X be an integrable random variable. We define **conditional expectation** of X under the event A as

$$\mathbb{E}[X \mid A] = \frac{\mathbb{E}[X\,\mathbb{1}_A]}{\mathbb{P}(A)} = \frac{1}{\mathbb{P}(A)} \times \begin{cases} \displaystyle\sum_{x \in X(A)} x\, p_X(x) \\ \displaystyle\int_{X(A)} x\, f_X(x)\, dx \end{cases} \tag{1.32}$$

This is a special case of conditional expectation with respect to an event, not with respect to a σ-algebra. Later, we generalize this notion to conditional expectation with respect to a σ-algebra.

Remark 1.17. In particular, if for an event $B \in \mathcal{F}$, we take $X = \mathbb{1}_B$, then

$$\mathbb{E}[X \mid A] = \frac{\mathbb{E}[\mathbb{1}_B \mathbb{1}_A]}{\mathbb{P}(A)} = \frac{\mathbb{E}[\mathbb{1}_{B \cap A}]}{\mathbb{P}(A)} = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)} = \mathbb{P}(B \mid A). \tag{1.33}$$

□

Remark 1.18. Especially in the case that X is discrete, we recall here that $X(A) = \{X(\omega) \colon \omega \in A\}$ and

$$p_X(x) = \mathbb{P}(X = x) = \mathbb{P}(\{\omega \in \Omega \colon X(\omega) = x\}). \tag{1.34}$$

Therefore, the right-hand side in (1.32) is

$$\begin{aligned} \sum_{x \in X(A)} x\, \frac{p_X(x)}{\mathbb{P}(A)} &= \sum_{x \in X(\Omega)} x\, \frac{\mathbb{P}(X = x)\, \mathbb{1}_{\{x \in X(A)\}}}{\mathbb{P}(A)} \\ &= \sum_{x \in X(\Omega)} x\, \frac{\mathbb{E}[\mathbb{1}_{\{X = x\}}\, \mathbb{1}_{\{x \in X(A)\}}]}{\mathbb{P}(A)} \\ &= \sum_{x \in X(\Omega)} x\, \frac{\mathbb{P}(\{X = x\} \cap \{x \in X(A)\})}{\mathbb{P}(A)} = \sum_{x \in X(\Omega)} x\, \mathbb{P}(X = x \mid A) \end{aligned} \tag{1.35}$$

which implies that the definition of conditional expectation is the expectation with respect to the conditional probability. □

1.4 Stochastic Convergence

1.4.1 Weak Convergence and Convergence in Distribution

Let μ, μ_n $(n = 1, 2, \cdots)$ be distributions[2)].

Definition 1.19. For all bounded, continuous function f, we say that random variables X_n converges to X in distribution if

$$\lim_{n \to \infty} \mathbb{E}[f(X_n)] = \mathbb{E}[f(X)] \tag{1.36}$$

The following is alternative definition of weak convergence.

[2)] Through this note we consider Borel probability measure on $\mathbb{R}$.

Definition 1.20. Let X be a real-valued continuous random variable with cumulative distribution function F_X and $(X_n)_{n\in\mathbb{N}}$ be a sequence of real-valued continuous random variables with F_{X_n}. The sequence (X_n) is said to **converge in distribution**, **converge in law** or **converge weakly** to X when

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x) \tag{1.37}$$

for all point $x \in \mathbb{R}$ where F_X is continuous. It is sometimes written like $X_n \xrightarrow{(\mathrm{d})} X$.

Remark 1.21. Actually the convergence in distribution of random variables is just a paraphrase of the weak convergence of measures because we have the following relation between random variable X and its measure ν. Let $F_\nu(x) \equiv \mathbb{P}_\nu(X \in (-\infty, x)$.

$$\mathbb{P}(X \in A) := \mathbb{P}_\nu(X \in A) = \int_A \frac{d\mathbb{P}_\nu}{d\mu}\, d\mu, \tag{1.38}$$

where μ is called a reference measure, especially in this case μ is the Lebesgue measure, and the Radon-Nikodym derivative $\frac{d\mathbb{P}_\nu}{d\mu} (= \frac{dF_\nu}{d\mu})$ is called *density*. □

Theorem 1.22. *A sequence of random variables $\{X_n\}$ converges to a random variable X in law if and only if characteristic function of X_n $\varphi_n(t)$ converges to that of X $\varphi(t)$ pointwisely and $\varphi(t)$ is continuous at $t = 0$.*

Remark 1.23. Since $|e^{i\theta x}| \le 1$,

$$\begin{aligned} \varphi_{X_n}(\theta) - \varphi_X(\theta) &= \int_{-\infty}^{\infty} e^{i\theta x}\, f_{X_n}(x)\, dx - \int_{-\infty}^{\infty} e^{i\theta x}\, f_X(x)\, dx \\ &= \int_{-\infty}^{\infty} e^{i\theta x}\, (f_{X_n}(x) - f_X(x))\, dx \le \int_{-\infty}^{\infty} (f_{X_n}(x) - f_X(x))\ dx \end{aligned} \tag{1.39}$$

The right-hand side goes to zero if X_n converges in distribution to X (i.e. $F_{X_n} \to F_X$). We contain some non-continuity points for F_X in the integral, but these points have weight zero under the Lebesgue measure. Therefore, the convergence in distribution implies the point-wise convergence of the characteristic functions. □

By Remark 1.23, to prove Theorem 1.22 it suffices to show that the convergence of the characteristic functions implies the convergence in distribution, which was called Glivenko's theorem[3)], which we will prove below.

Theorem 1.24 (Glivenko)**.** *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables and X be a random variable with characteristic functions φ_n and φ, respectively. If for any $\theta \in \mathbb{R}$,*

$$\lim_{n\to\infty} \varphi_{X_n}(\theta) = \varphi_X(\theta) \tag{1.40}$$

then X_n converges to X in law.

[3)] We sometimes call Theorem 1.22 Levy's continuity theorem, which states that the weak convergence of the measure is equivalent to the point-wise convergence of the characteristic functions:

(**Levy's continuity theorem**) *Let $(\mu_n)_{n\in\mathbb{N}}$ be a sequence of the probability measures with the characteristic function $\varphi_n = \varphi_{\mu_n}$. If φ_n converges point-wise to some φ and converges uniformly near $\theta = 0$, then μ_n also converges weakly to some μ and φ is the characteristic function of μ.*

Proof. Let μ_n and μ be density function of X_n and X, respectively. We split the proof into several steps:

Step 1. For bounded (integrable) function g on $\mathbb{R}$, we set

$$h(x) = \int_{-\infty}^{\infty} e^{i\theta x} g(\theta)\, d\theta \tag{1.41}$$

Then we prove that

$$\int_{-\infty}^{\infty} h(x)\, \mu_n(dx) = \int_{-\infty}^{\infty} h(x)\, \mu_(dx) \tag{1.42}$$

Since by Fubini theorem,

$$\begin{aligned} \int_{-\infty}^{\infty} h(x)\, \mu_n(dx) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{i\theta x} g(\theta)\, d\theta\, \mu_n(dx) \\ &= \int_{-\infty}^{\infty} \varphi_n(\theta)\, g(\theta)\, d\theta \end{aligned} \tag{1.43}$$

We note that the integrands in both sides are bounded (integrable). Then we apply the dominated convergence theorem and obtain

$$\begin{aligned} \lim_{n\to\infty} \int_{-\infty}^{\infty} h(x)\, \mu_n(dx) &= \lim_{n\to\infty} \int_{-\infty}^{\infty} \varphi_n(\theta)\, g(\theta)\, d\theta \\ &= \int_{-\infty}^{\infty} \lim_{n\to\infty} \varphi_n(\theta)\, g(\theta)\, d\theta \\ &= \int_{-\infty}^{\infty} \varphi(\theta)\, g(\theta)\, d\theta \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{i\theta x} g(\theta)\, \mu(dx)\, d\theta \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{i\theta x} g(\theta)\, d\theta\, \mu(dx) \\ &= \int_{-\infty}^{\infty} h(x)\, \mu(dx) \end{aligned} \tag{1.44}$$

Step 2. If a continuous function f which has a compact support can be approximated by h, then by Step 1, we have

$$\int_{-\infty}^{\infty} f(x)\, \mu_n(dx) \xrightarrow[n\to\infty]{} \int_{-\infty}^{\infty} f(x)\, \mu(dx), \tag{1.45}$$

which conclude the convergence in law. Thus we prove that f can be approximated by h. Set

$$g_n(z) := \frac{1}{2\pi} e^{-\frac{z^2}{2n}} \int_{-\infty}^{\infty} e^{-izy} f(y)\, dy \tag{1.46}$$

Then since f is continuous and has a compact support (therefore it has the maximum),

$$|g_n(z)| \le \frac{1}{2\pi} e^{-\frac{z^2}{2n}} \int_{-\infty}^{\infty} |f(y)|\, dy < \infty, \tag{1.47}$$

which implies that g_n is integrable. By computing the characteristic function of normal distri-

bution, we get

$$\begin{aligned} h_n(x) &:= \int_{-\infty}^{\infty} e^{izx} g_n(z)\, dz \\ &= \int_{-\infty}^{\infty} e^{izx} \frac{1}{2\pi} e^{-\frac{z^2}{2n}} \int_{-\infty}^{\infty} e^{-izy} f(y)\, dy\, dz \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{iz(x-y)}\, e^{-\frac{z^2}{2n}}\, dz\, f(y)\, dy \\ &= \sqrt{\frac{n}{2\pi}} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} e^{-z(x-y)} \frac{1}{\sqrt{2\pi n}} e^{-\frac{z^2}{2n}}\, dz \right) f(y)\, dy \\ &= \sqrt{\frac{n}{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{n(x-y)^2}{2}} f(y)\, dy \\ &= \sqrt{\frac{1}{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} f\left(x + \frac{t}{\sqrt{n}}\right) dt \end{aligned} \tag{1.48}$$

Therefore, again by the dominated convergence theorem,

$$\lim_{n\to\infty} h_n(x) = \sqrt{\frac{1}{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} \lim_{n\to\infty} f\left(x + \frac{t}{\sqrt{n}}\right) dt = \sqrt{\frac{1}{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} f(x)\, dt = f(x) \tag{1.49}$$

which yields the pointwise convergence.

Step 3. We prove the uniform convergence $h_n \to f$. Since

$$\begin{aligned} |h_n(x) - f(x)| &= \left| \sqrt{\frac{1}{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} f\left(x + \frac{t}{\sqrt{n}}\right) dt - \sqrt{\frac{1}{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} f(x)\, dt \right| \\ &\le \sqrt{\frac{1}{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} \left| f\left(x + \frac{t}{\sqrt{n}}\right) - f(x) \right| dt \end{aligned} \tag{1.50}$$

for any $a \in \mathbb{R}$, we separate the range of integral into $|t| \le a$ and $|t| > a$. For $|t| \le a$,

$$\begin{aligned} &\sup_{x\in\mathbb{R}} \int_{|t|\le a} e^{-\frac{t^2}{2}} \left| f\left(x + \frac{t}{\sqrt{n}}\right) - f(x) \right| dt \\ &\qquad \le \sup_{|t|\le a,\, x\in\mathbb{R}} \left| f\left(x + \frac{t}{\sqrt{n}}\right) - f(x) \right| \int_{|t|\le a} e^{-\frac{t^2}{2}}\, dt \xrightarrow[n\to\infty]{} 0 \end{aligned} \tag{1.51}$$

On the other hand, for $|t| > a$,

$$\sup_{x\in\mathbb{R}} \int_{|t|>a} e^{-\frac{t^2}{2}} \left| f\left(x + \frac{t}{\sqrt{n}}\right) - f(x) \right| dt \le 2 \sup_{x\in\mathbb{R}} |f(x)| \int_{|t|>a} e^{-\frac{t^2}{2}}\, dt \le M \int_{|t|>a} e^{-\frac{t^2}{2}}\, dt, \tag{1.52}$$

where the constant M does not depend on n. Therefore, we have

$$\limsup_{n\to\infty} \left(\sup_{x\in\mathbb{R}} |h_n(x) - f(x)| \right) \le \frac{M}{\sqrt{2\pi}} \int_{|t|>a} e^{-\frac{t^2}{2}}\, dt \tag{1.53}$$

Since the left-hand side does not depend on a, taking $a \to \infty$, we obtain

$$\int_{|t|>a} e^{-\frac{t^2}{2}}\, dt \xrightarrow[a\to\infty]{} 0, \tag{1.54}$$

which gives that $h_n(x)$ converges uniformly to $f(x)$. ■

Question 1.25. Let $X_1, X_2, \cdots$ be i.i.d. random variables distributed by Poisson with parameter 1. We set

$$Y_n = \sum_{k=1}^{n} X_k, \qquad S_n = \frac{Y_n - \mathbb{E}[Y_n]}{\mathbb{V}\mathrm{ar}(Y_n)} \tag{1.55}$$

Then S_n converges to $N(0,1)$ in distribution.

1.4.2 Convergence in Probability

From now on, let X and $(X_n)_{n\in\mathbb{N}}$ be a real-valued random variable and a sequence of real-valued random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, respectively.

Definition 1.26. The sequence (X_n) is said **convergent in probability** to X when for any $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0. \tag{1.56}$$

We denote the convergence in probability by

$$X_n \underset{n\to\infty}{\to} X \qquad \text{in probability.} \tag{1.57}$$

It is sometimes written like $X_n \overset{\text{prob.}}{\longrightarrow} X$.

By the continuous mapping theorem, for any continuous function ψ, if $X_n \overset{\text{prob.}}{\longrightarrow} X$, then $\psi(X_n) \overset{\text{rob.p}}{\longrightarrow} \psi(X)$.

Proposition 1.27. (**Markov's inequality and Chebyshev's inequality**) *Let ψ be a non-negative measurable function. Then, we have that for any $a > 0$,*

$$\mathbb{P}(\psi(X) \geq a) \leq \frac{\mathbb{E}[\psi(X)]}{a} \tag{1.58}$$

In particular, for the case that $\psi(X) = (X - \mathbb{E}[X])^2$ and $\sqrt{a} = \varepsilon$, the inequality is called Chebyshev's inequality:

$$\mathbb{P}\left(|X - \mathbb{E}[X]| \geq \varepsilon\right) \leq \frac{\mathbb{V}ar(X)}{\varepsilon^2} \tag{1.59}$$

Proof. We divide the expectation as

$$\begin{aligned}
\mathbb{E}[\psi(X)] &= \mathbb{E}[\psi(X)\mathbb{1}_{\{\psi(X)\geq a\}}] + \mathbb{E}[\psi(X)\mathbb{1}_{\{\psi(X)<a\}}] \\
&\geq \mathbb{E}[\psi(X)\mathbb{1}_{\{\psi(X)\geq a\}}] \\
&\geq \mathbb{E}[a\,\mathbb{1}_{\{\psi(X)\geq a\}}] \\
&= a\,\mathbb{P}(\psi(X) \geq a),
\end{aligned} \tag{1.60}$$

which yields Markov's inequality. For Chebyshev's inequality, we have that

$$\mathbb{P}\left((X - \mathbb{E}[X])^2 \geq \varepsilon^2\right) = \mathbb{P}\left(|X - \mathbb{E}[X]| \geq \varepsilon\right) \leq \frac{\mathbb{E}[(X - \mathbb{E}[X])^2]}{\varepsilon^2} = \frac{\mathbb{V}\text{ar}(X)}{\varepsilon^2}, \tag{1.61}$$

which completes the proof. ■

Theorem 1.28. *Assume that X has a density f_X. Then*

$$X_n \overset{\text{prob.}}{\longrightarrow} X \quad \Longrightarrow \quad X_n \overset{\text{(d)}}{\longrightarrow} X. \tag{1.62}$$

Proof. Let F_X and F_{X_n} be the distribution functions of X and X_n respectively. Let $x \in \mathbb{R}$ be a continuity point of F_X. Then, we have that for any $\varepsilon > 0$, there exists a $\delta = \delta(\varepsilon, x) > 0$ such that

$$\int_{x-\delta}^{x+\delta} f_X(y)\, dy \;\leq\; \frac{\varepsilon}{2}, \tag{1.63}$$

where f_X is the density function of X. To prove the convergence of X_n in distribution to X, it suffices to show that for any $x \in \mathbb{R}$, there exists an $N = N(\varepsilon, \delta) \in \mathbb{N}$ such that for $n \geq N$,

$$|F_{X_n}(x) - F_X(x)| \leq \varepsilon. \tag{1.64}$$

For any $x \in \mathbb{R}$, we have that

$$\begin{aligned} |F_{X_n}(x) - F_X(x)| &= |\mathbb{P}(X_n \leq x) - \mathbb{P}(X \leq x)| \\ &= |\mathbb{P}(X_n \leq x, X \leq x) + \mathbb{P}(X_n \leq x, X > x) \\ &\qquad - \{\mathbb{P}(X \leq x, X_n \leq x) + \mathbb{P}(X \leq x, X_n > x)\}| \\ &= |\mathbb{P}(X_n \leq x < X) - \mathbb{P}(X \leq x < X_n)| \\ &\leq \mathbb{P}(X_n \leq x < X) + \mathbb{P}(X \leq x < X_n). \end{aligned} \tag{1.65}$$

For the first term, we further divide it into two parts:

$$\mathbb{P}(X_n \leq x < X) = \mathbb{P}(X_n \leq x < X,\, |X - X_n| < \delta) + \mathbb{P}(X_n \leq x < X,\, |X - X_n| \geq \delta). \tag{1.66}$$

Since

$$X_n \leq x < X,\ |X - X_n| < \delta \quad \Rightarrow \quad x < X < x + \delta, \tag{1.67}$$

we have that

$$\mathbb{P}(X_n \leq x < X,\, |X - X_n| < \delta) \leq \mathbb{P}(x < X < x + \delta) = \int_x^{x+\delta} f_X(y)\, dy. \tag{1.68}$$

Since

$$\mathbb{P}(X_n \leq x < X,\, |X - X_n| \geq \delta) \leq \mathbb{P}(|X - X_n| \geq \delta), \tag{1.69}$$

combining with (1.68), for the first term in (1.65) we obtain that

$$\mathbb{P}(X_n \leq x < X) \leq \int_x^{x+\delta} f_X(y)\, dy + \mathbb{P}(|X - X_n| \geq \delta). \tag{1.70}$$

For the second term in (1.65), similarly to the first term, since

$$X \leq x < X_n,\ |X - X_n| < \delta \quad \Rightarrow \quad x - \delta < X \leq x, \tag{1.71}$$

we have that

$$\mathbb{P}(X \leq x < X_n,\, |X - X_n| < \delta) \leq \mathbb{P}(x - \delta < X \leq x) = \int_{x-\delta}^{x} f_X(y)\, dy. \tag{1.72}$$

Therefore, we obtain that

$$\mathbb{P}(X \leq x < X_n) \leq \int_{x-\delta}^{x} f_X(y)\, dy + \mathbb{P}(|X - X_n| \geq \delta). \tag{1.73}$$

Substituting (1.70) and (1.73) into (1.65), we have that

$$|F_{X_n}(x) - F_X(x)| \leq 2\, \mathbb{P}(|X - X_n| \geq \delta) + \int_{x-\delta}^{x+\delta} f_X(y)\, dy. \tag{1.74}$$

Since X_n converges in probability to X, we have that for any $\varepsilon > 0$, there exists an $N = N(\varepsilon, \delta) \in \mathbb{N}$ such that for any $n \geq N$,

$$\mathbb{P}(|X_n - X| \geq \delta) \leq \frac{\varepsilon}{4}. \tag{1.75}$$

Hence, we obtain that, by combining with (1.63), $|F_{X_n}(x) - F_X(x)| \leq \varepsilon$, which completes the proof. ■

The opposite statement "convergence in distribution implies convergence in probability" is held only when the limiting random variable X is a constant.

Theorem 1.29. *For some constant c, we have*

$$X_n \xrightarrow{\text{(d)}} c \quad \Longrightarrow \quad X_n \xrightarrow{\text{prob.}} c. \tag{1.76}$$

Proof. Without loss of generality, we take $c = 0$. Let g be a continuous function satisfying

$$g(x) = \begin{cases} \frac{1}{\varepsilon}\, x + 2 & x \in [-2\varepsilon, -\varepsilon] \\ 0 & x \in [-\varepsilon, \varepsilon] \\ \frac{1}{\varepsilon}\, x - 1 & x \in [\varepsilon, 2\varepsilon] \\ 1 & x \in (-\infty, -2\varepsilon] \cup [2\varepsilon, \infty). \end{cases} \tag{1.77}$$

Then, since g is continuous and X_n converges in distribution to 0,

$$\mathbb{E}[g(X_n)] \to \mathbb{E}[g(0)] = 0. \tag{1.78}$$

Therefore, since by the definition of g,

$$\mathbb{P}(|X_n| \geq 2\varepsilon) = \mathbb{E}[\mathbb{1}_{\{|X_n| \geq 2\varepsilon\}}] \leq \mathbb{E}[g(X_n)], \tag{1.79}$$

we have that

$$\mathbb{P}(|X_n| \geq 2\varepsilon) \underset{n\to\infty}{\longrightarrow} 0, \tag{1.80}$$

which implies the convergence of X_n in probability to $c = 0$. ■

Question 1.30 (Weak law of large numbers)**.** Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of $\mathbb{R}$-valued i.i.d. random variables. We assume that

$$\mathbb{E}[X_1] = \mu, \qquad \mathbb{V}\mathrm{ar}(X_1) = \sigma^2 < \infty. \tag{1.81}$$

Then, show that

$$\frac{1}{n}\sum_{k=1}^{n} X_k \xrightarrow[n\to\infty]{p} \mu. \tag{1.82}$$

1.4.3 Almost Sure Convergence

Same as the previous sections, let X and $(X_n)_{n\in\mathbb{N}}$ be a real-valued random variable and a sequence of real-valued random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, respectively.

Definition 1.31. The sequence (X_n) is said to **almost surely converge** to a random variable X when for any $\varepsilon > 0$,

$$\mathbb{P}\left(|X_n - X| > \varepsilon \ \text{ infinitely often }\right) = 0. \tag{1.83}$$

We denote the almost sure convergence by

$$X_n \underset{n\to\infty}{\to} X \qquad \text{a.s..} \tag{1.84}$$

It is sometimes written like $X_n \xrightarrow{\text{a.s.}} X$.

Remark 1.32. The definition (1.83) is equivalent to

$$\mathbb{P}\left(\lim_{n\to\infty} X_n = X\right) = 1. \tag{1.85}$$

It is quite important to understand the difference between definitions of convergence in probability and almost sure convergence. □

Definition 1.33. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(A_n)_{n\in\mathbb{N}}$ be a sequence of events. We define the **limit supremum** of the sequence of event

$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k. \tag{1.86}$$

We also define the **limit infimum** of the sequence of event

$$\liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k. \tag{1.87}$$

We can translate

$$\omega \in \limsup_{n\to\infty} A_n = \text{there exist infinitely many } n_k = n_k(\omega) \text{ such that } \omega \in A_{n_k}$$

We also say that "A_n **occurs infinitely often**" when $\bigcap_{n\geq 1} \bigcup_{k\geq n} A_k$ occurs. We denote the limit supremum of the sequence of event $\limsup_{n\to\infty} A_n$ by

$$\{A_n \quad \text{i.o.}\}. \tag{1.88}$$

For the limit infimum of the sequence of event, we can say that "all A_n except for finite number of events occur" when $\bigcup_{n\geq 1} \bigcap_{k\geq n} A_k$ occurs.

Lemma 1.34. (**Fatou-Lebesgue theorem**) *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(A_n)_{n\in\mathbb{N}}$ be a sequence of events. Then, we have*

$$\mathbb{P}\left(\liminf_{n\to\infty} A_n\right) \leq \liminf_{n\to\infty} \mathbb{P}(A_n) \leq \limsup_{n\to\infty} \mathbb{P}(A_n) \leq \mathbb{P}\left(\limsup_{n\to\infty} A_n\right). \tag{1.89}$$

Proof. We will prove only for the limit infimum. By the definition, we have that

$$\mathbb{P}\left(\liminf_{n\to\infty} A_n\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k\right). \tag{1.90}$$

Since the event $E_n = \bigcap_{k=n}^{\infty} A_k$ is increasing (i.e. $E_1 \subset E_2 \subset \cdots$), by the continuity, we have that

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k\right) = \lim_{n\to\infty} \mathbb{P}\left(\bigcap_{k=n}^{\infty} A_k\right). \tag{1.91}$$

Since for any $k \geq n$,

$$\mathbb{P}\left(\bigcap_{k=n}^{\infty} A_k\right) \leq \mathbb{P}(A_k), \tag{1.92}$$

we have that

$$\mathbb{P}\left(\bigcap_{k=n}^{\infty} A_k\right) \leq \inf_{k\leq n} \mathbb{P}(A_k). \tag{1.93}$$

Therefore, we conclude that

$$\lim_{n\to\infty} \mathbb{P}\left(\bigcap_{k=n}^{\infty} A_k\right) \leq \lim_{n\to\infty} \inf_{k\leq n} \mathbb{P}(A_k), \tag{1.94}$$

which completes the proof of the first inequality. ■

Remark 1.35. The Fatou-Lebesgue theorem is usually stated in measure theory as a chain of inequalities relating the integral, limit supremum and limit infimum. We can replace $\mathbb{P}$ and A_n to $\mathbb{E}$ and X_n. In the case we consider random variables X_n, if the sequence of random variables converges point-wise, then the inequalities turn into equalities, which implies that the theorem reduces to the (Lebesgue's) **dominated convergence theorem**. □

Question 1.36. Show the following equivalence: for any $\varepsilon > 0$,

$$\mathbb{P}\left(\lim_{n\to\infty} |X_n - X| > \varepsilon\right) = 0 \quad \Longleftrightarrow \quad \mathbb{P}\left(\limsup_{n\to\infty} \{|X_n - X| > \varepsilon\}\right) = 0. \tag{1.95}$$

By Question 1.36, we can say that

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} \{|X_k - X| \geq \varepsilon\}\right) = 0 \quad \Longleftrightarrow \quad X_n \xrightarrow{\text{a.s.}} X. \tag{1.96}$$

Theorem 1.37.

$$X_n \xrightarrow{a.s.} X \quad \Longrightarrow \quad X_n \xrightarrow{prob.} X. \tag{1.97}$$

Proof. For any $\varepsilon > 0$, let

$$A_n = \bigcup_{k=n}^{\infty} \{|X_k - X| \geq \varepsilon\}. \tag{1.98}$$

Then, the sequence of the event (A_n) is decreasing. Therefore, by the continuity of the probability, we have that

$$\mathbb{P}(\bigcap_{n=1}^{\infty} A_n) = \lim_{n\to\infty} \mathbb{P}(A_n), \tag{1.99}$$

which implies that

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} \{|X_k - X| \geq \varepsilon\}\right) = \lim_{n\to\infty} \mathbb{P}\left(\bigcup_{k=n}^{\infty} \{|X_k - X| \geq \varepsilon\}\right). \tag{1.100}$$

Since

$$\mathbb{P}\left(\bigcup_{k=n}^{\infty} \{|X_k - X| \geq \varepsilon\}\right) \geq \mathbb{P}\left(|X_n - X| \geq \varepsilon\right) \geq 0, \tag{1.101}$$

we obtain that

$$\lim_{n\to\infty} \mathbb{P}\left(|X_n - X| \geq \varepsilon\right) = 0, \tag{1.102}$$

which implies that $X_n \xrightarrow{\text{prob.}} X$. ■

The opposite statement "convergence in probability implies almost sure convergence" is held for a proper subsequence of (X_n).

Theorem 1.38. *If (X_n) converges in probability to X, i.e., $X_n \xrightarrow{prob.} X$, there exists a subsequence $(X_{n_i})_{i\in\mathbb{N}}$ such that as $i \to \infty$,*

$$X_{n_i} \xrightarrow{a.s.} X. \tag{1.103}$$

Proof. Let $\{n_i\}_{i\in\mathbb{N}}$ be a subsequence satisfying $n_1 < n_2 < \cdots$ and

$$\mathbb{P}\left(|X_{n_k} - X| > 2^{-k}\right) < 2^{-k}. \tag{1.104}$$

By the Borel-Cantelli lemma,

For the sum of independent real-valued random variables, almost sure convergence is equivalent to convergence in probability.

Theorem 1.39. *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of* independent *real-valued random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and*

$$S_n = \sum_{k=1}^{n} X_k \tag{1.105}$$

be a sum of X_n. Then we have

$$S_n \xrightarrow{\text{a.s.}} X \quad \Longleftrightarrow \quad S_n \xrightarrow{\text{prob.}} X. \tag{1.106}$$

For $\varepsilon > 0$, the sequence (X_n) is *converges almost completely* to a random variable X when

$$\sum_{n\in\mathbb{N}} \mathbb{P}\left(|X_n - X| > \varepsilon\right) < \infty. \tag{1.107}$$

This is a key for the **Borel-Cantelli lemma**, which is the most basic theorem to show the almost sure convergence.

Theorem 1.40. *If the sequence (X_n) converges almost completely to X, then*

$$X_n \xrightarrow{\text{a.s.}} X. \tag{1.108}$$

Theorem 1.41. (**Borel-Cantelli lemma**) *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.*

(The first Borel-Cantelli lemma):

$$\sum_{n\in\mathbb{N}} \mathbb{P}\left(A_n\right) < \infty \quad \Longrightarrow \quad \mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = 0. \tag{1.109}$$

(The second Borel-Cantelli lemma): *Assume that A_n are independent. Then,*

$$\sum_{n\in\mathbb{N}} \mathbb{P}\left(A_n\right) = \infty \quad \Longrightarrow \quad \mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = 1. \tag{1.110}$$

Proof. (The first Borel-Cantelli lemma): By the definition,

$$\mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = \mathbb{P}\left(\bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} A_k\right). \tag{1.111}$$

Since the event $\bigcup_{k=n}^{\infty} A_k$ is decreasing, by the continuity, we have that

$$\begin{aligned}\mathbb{P}\left(\bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} A_k\right) &= \lim_{n\to\infty}\mathbb{P}\left(\bigcup_{k=n}^{\infty} A_k\right)\\ &\leq \lim_{n\to\infty}\sum_{k=n}^{\infty}\mathbb{P}(A_k) = \lim_{n\to\infty}\left(\sum_{k=1}^{\infty}\mathbb{P}(A_k) - \sum_{k=1}^{n-1}\mathbb{P}(A_k)\right),\end{aligned} \tag{1.112}$$

where the inequality is due to the sub-additivity (Boole's inequality). Therefore, by the assumption, we conclude the statement.

(The second Borel-Cantelli lemma): By De Morgan's law, it is sufficient to show that

$$\mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = 1 \iff \mathbb{P}\left(\bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty} A_k^c\right) = 0. \tag{1.113}$$

Since the event $\bigcap_{k=n}^{\infty} A_k^c$ is increasing, by the continuity, we have that

$$\begin{aligned}\mathbb{P}\left(\bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty} A_k^c\right) &= \lim_{n\to\infty}\mathbb{P}\left(\bigcap_{k=n}^{\infty} A_k^c\right)\\ &= \lim_{n\to\infty}\prod_{k=n}^{\infty}\mathbb{P}(A_k^c)\\ &= \lim_{n\to\infty}\prod_{k=n}^{\infty}(1-\mathbb{P}(A_k))\\ &\leq \lim_{n\to\infty}\prod_{k=n}^{\infty}\exp\{-\mathbb{P}(A_k)\}\\ &= \lim_{n\to\infty}\exp\left\{-\sum_{k=n}^{\infty}\mathbb{P}(A_k)\right\} = 0,\end{aligned} \tag{1.114}$$

where the second equality is due to independence, the inequality comes from

$$0 \leq 1 - x \leq e^{-x} \qquad (0 \leq x \leq 1) \tag{1.115}$$

and the last equality is by the assumption. We note here that by the assumption,

$$\sum_{k=n}^{\infty}\mathbb{P}(A_k) = \lim_{m\to\infty}\sum_{k=n}^{n+m}\mathbb{P}(A_k) = \infty. \tag{1.116}$$

Hence we conclude the statement. ■

Question 1.42 (Strong law of large numbers)**.** Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of $\mathbb{R}$-valued i.i.d. random variables. We assume that

$$\mathbb{E}[X_1] = \mu, \qquad \mathbb{E}[X_1^4] < \infty. \tag{1.117}$$

Then, show that

$$\frac{1}{n}\sum_{k=1}^{n} X_k \xrightarrow[n\to\infty]{a.s.} \mu. \tag{1.118}$$

1.4.4 L^p-Convergence

Same as before, let X and $(X_n)_{n\in\mathbb{N}}$ be a real-valued random variable and a sequence of real-valued random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, respectively.

Definition 1.43. For $p > 0$, the sequence (X_n) is said to **L^p-converge** to a random variable X when

$$\lim_{n\to\infty} \mathbb{E}\left[|X_n - X|^p\right] = 0. \tag{1.119}$$

We denote the L^p-convergence by

$$X_n \underset{n\to\infty}{\to} X \qquad \text{in } L^p. \tag{1.120}$$

It is sometimes written like $X_n \xrightarrow{L^p} X$.

Remark 1.44. Especially for the case $p = 1$, we call convergence in mean:

$$\mathbb{E}\left[|X_n - X|\right] = 0 \quad \Rightarrow \quad \lim_{n\to\infty} \mathbb{E}[|X_n|] = \mathbb{E}[|X|]. \tag{1.121}$$

Also for the case $p = 2$, we call convergence in quadratic mean (or in mean square). □

Theorem 1.45. *For any $p \geq 1$, we have that*

$$X_n \xrightarrow{L^p} X \quad \Longrightarrow \quad X_n \xrightarrow{\text{prob.}} X. \tag{1.122}$$

1.4.5 Uniform integrability

Definition 1.46. The sequence (X_n) is said to **uniformly integrable** when

$$\lim_{a\to\infty} \sup_{n\in\mathbb{N}} \mathbb{E}\left[|X_n| \mathbb{1}_{\{|X_n|>a\}}\right] = 0. \tag{1.123}$$

Theorem 1.47. *The sequence (X_n) converges in L^1 to X if and only if (X_n) is uniformly integrable and converges in probability to X.*

A Convergence in distribution

We proved Glivenko's theorem in the main part of the note. In this appendix we consider the convergence in distribution more detailed. We first see the generalized formula of integration by part:

$$\int_a^b f'(x)F(x)dx = [f(x)F(x)]_a^b - \int_a^b f(x)\,\mu(dx), \tag{A.124}$$

where F is the distribution function and μ is the measure of f.

Theorem A.1. *Let φ_n, φ be the characteristic functions of μ_n and μ, respectively. Then we have*

$$\mu_n \to \mu \quad \Rightarrow \quad \varphi_n(t) \to \varphi(t) \tag{A.125}$$

uniformly on compact sets.

Proof. Let F_n, F be the distribution function of μ_n and μ, respectively. Since $\mu_n \to \mu$,

$$F_n(x) \to F(x) \tag{A.126}$$

Since for any $\varepsilon > 0$ and continuous point a ($-a$ also) large enough, we have $F(a) - F(-a) > 1 - \varepsilon$, there is $N_1 = N_1(\varepsilon)$ such that for $n > N_1$,

$$F_n(a) - F_n(-a) > 1 - \varepsilon \tag{A.127}$$

Thus for $n > N_1$,

$$\begin{aligned}
\left| \varphi_n(t) - \int_{(-a,a]} e^{izx}\, \mu_n(dx) \right| &= \left| \int_{-\infty}^{\infty} e^{izx}\, \mu_n(dx) - \int_{(-a,a]} e^{izx}\, \mu_n(dx) \right| \\
&= \left| \int_{(-a,a]^c} e^{izx}\, \mu_n(dx) \right| \\
&\le 1 - \mu_n((-a,a]) \\
&= 1 - \Big(F_n(a) - F_n(-a) \Big) < \varepsilon
\end{aligned} \tag{A.128}$$

Similarly we get $\left| \varphi(t) - \int_{(-a,a]} e^{izx}\, \mu(dx) \right| < \varepsilon$. From the generalized integration by parts, we have

$$\begin{aligned}
\left| \int_{(-a,a]} e^{izx}\, \mu_n(dx) - \int_{(-a,a]} e^{izx}\, \mu_(dx) \right| &= \Big| e^{iza}(F_n(a) - F(a)) + e^{-iza}(F_n(-a) - F(-a)) \\
&\qquad + \int_{(-a,a]} ize^{izx}\, (F_n(x) - F(x))\, dx \Big| \\
&\le \Big| e^{iza}(F_n(a) - F(a)) \Big| + \Big| e^{-iza}(F_n(-a) - F(-a)) \Big| \\
&\qquad + \left| \int_{(-a,a]} ize^{izx}\, (F_n(x) - F(x))\, dx \right| \\
&\le |F_n(a) - F(a)| + |F_n(-a) - F(-a)| \\
&\qquad + \int_{(-a,a]} |z|\, |F_n(x) - F(x)|\, dx
\end{aligned} \tag{A.129}$$

Since the number of discontinuous points are finite, the right-hand side goes to 0. Therefore, by triangle inequality, for a compact set K, we obtain

$$\limsup_{n\to\infty} \sup_{t\in K} |\varphi_n(t) - \varphi(t)| \le 2\varepsilon, \tag{A.130}$$

which yields the convergence of the characteristic function. ■

For the characteristic functions φ_n, if we have $\varphi_n \to \varphi$, we cannot conclude $\mu_n \to \mu$ directly. Even if $\varphi_n \to \varphi$, it might occur that φ is not the characteristic function. The following theorem yields the condition so that such situation does not occur.

Theorem A.2 (Levy's continuity theorem)**.** *Let $(\mu_n)_{n\in\mathbb{N}}$ be a sequence of the probability measures with the characteristic function $\varphi_n = \varphi_{\mu_n}$. If φ_n converges point-wisely to some φ and converges uniformly near $\theta = 0$, then μ_n also converges weakly to some μ and φ is the characteristic function of μ.*

Proof. First we fix $a > 0$ and consider $|t| < a$. Since $|e^{itx}| < 1$, by the dominated convergence theorem,

$$t_n \to t \quad \Rightarrow \quad \varphi(t_n) \to \varphi(t), \tag{A.131}$$

which tells the continuity of the characteristic functions. By the argument in Glivenko's theorem, the convergence $\varphi_n \to \varphi$ is uniform in $|t| < a$. Thus $\varphi(t)$ is continuous at $t = 0$ and

$$\varphi(0) = \lim_{n\to\infty} \varphi_n(0) = 1, \tag{A.132}$$

which is equivalent to $\forall \varepsilon > 0, \exists\, \delta = \delta(\varepsilon) < a, \quad |t| < \delta \ \Rightarrow\ |\varphi(t) - 1| < \varepsilon$. Since φ_n converges uniformly to φ,

$$\forall \varepsilon > 0, \exists\, N = N(\varepsilon) a, \quad n > N, |t| < \delta \ \Rightarrow\ |\varphi(t) - \varphi_n(t)| < \varepsilon \tag{A.133}$$

By the triangle inequality, we obtain

$$\forall \varepsilon > 0, \exists\, N = N(\varepsilon) a, \quad n > N, |t| < \delta \ \Rightarrow\ |\varphi_n(t) - 1| < \varepsilon \tag{A.134}$$

Then

$$\begin{aligned} 1 - \varepsilon &\le \frac{1}{2\delta} \int_{-\delta}^{\delta} \mathcal{R}(\varphi_n(t))\, dt \\ &= \frac{1}{2\delta} \int_{-\delta}^{\delta} \int_{-\infty}^{\infty} \cos tx\ \mu_n(dx)\, dt \\ &= \int_{-\infty}^{\infty} \frac{\sin \delta x}{\delta x} \mu_n(dx) \\ &= \int_{[-c,c]} \frac{\sin \delta x}{\delta x} \mu_n(dx) + \int_{[-c,c]^c} \frac{\sin \delta x}{\delta x} \mu_n(dx) \\ &\le \mu_n([-c,c]) + \frac{1}{\delta c}(1 - \mu_n([-c,c])) \\ &\le \mu_n([-c,c]) + \frac{1}{\delta c} \end{aligned} \tag{A.135}$$

Therefore, we obtain

$$\lim_{c\to\infty} \inf_{n>N} \mu_n([-c,c]) \ge 1 - \varepsilon \tag{A.136}$$

On the other hand, for each n, we have $\mu_n([-c,c]) \to 1$ as $c \to \infty$. Therefore,

$$\lim_{c\to\infty} \inf_{n\le N} \mu_n([-c,c]) = 1 \ge 1 - \varepsilon, \tag{A.137}$$

which implies that

$$\lim_{c\to\infty} \inf_{n} \mu_n([-c,c]) \ge 1 - \varepsilon \xrightarrow[\varepsilon\to 0]{} 1 \tag{A.138}$$

Hence $\{\mu_n\}$ has a convergent subsequence $\{\nu_n\}$ such that $\nu_n \to \mu$ as $n \to \infty$. We then have $\varphi_{\nu_n} \to \varphi_\mu$ point-wisely. Since φ_{ν_n} is a subsequence of φ_n, $\varphi_n \to \varphi$. Thus we conclude $\varphi = \varphi_\mu$. ■